form of one-cup-size amounts. That might make sense when we're talking about matter particles like electrons and protons, but just a few decades earlier Maxwell had triumphantly explained that light was a wave, not a particle. Einstein's proposal was threatening to undo that triumph. Planck himself was reluctant to accept this wild new idea, but it did explain the data. In a wild new idea's search for acceptance, that's a powerful advantage to have.

Meanwhile another problem was lurking over on the particle side of the ledger, where Rutherford's model explained atoms in terms of electrons orbiting nuclei.

Remember that if you shake an electron, it emits light. By "shake" we just mean accelerate in some way. An electron that does anything other than move in a straight line at a constant velocity should emit light.

From the picture of the Rutherford atom, with electrons orbiting around the nucleus, it certainly looks like those electrons are not moving in straight lines. They're moving in circles or ellipses. In a classical world, that unambiguously means that the electrons are being accelerated, and equally unambiguously that they should be giving off light. Every single atom in your body, and in the environment around you, should be glowing, if classical mechanics was right. That means the electrons should be losing energy as they emit radiation, which in turn implies that they should spiral downward into the central nucleus. Classically, electron orbits should not be stable.